```
Main [Java Application] C:\Users\sujit\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.8.v20250724-1412\jre\bin\javaw.exe  (Oct 16,

========== SMARTLIBRARY ==========

1 Dashboard and Statistics
2 Book Management
3 Smart Pattern Search
4 Intelligent Reading Planner
5 Borrowing and Resource Allocation
6 NP-Completeness and Optimization Lab
7 Randomized Algorithms
8 Parallel Algorithms
9 Algorithm Performance Comparison
10 Save Data
0 Exit
Select option: 1
Titles: 4 | Copies: 9 | Available: 9 | Active loans: 0 | Reservations: 0
Popular books:
B001  Algorithms                  Thomas Cormen        9780262518802  Algorithms       4/4
B002  Clean Code                   Robert C. Martin     9780132350884  Programming      2/2
B003  Introduction to Machine Lea… Ethem Alpaydin       9780262028189  AI               2/2
B004  Operating System Concepts    Silberschatz         9781119800361  Systems          1/1
```

```
1 Dashboard and Statistics
2 Book Management
3 Smart Pattern Search
4 Intelligent Reading Planner
5 Borrowing and Resource Allocation
6 NP-Completeness and Optimization Lab
7 Randomized Algorithms
8 Parallel Algorithms
9 Algorithm Performance Comparison
10 Save Data
0 Exit
Select option: 3
Pattern: rabin karp
1 KMP 2 Z 3 Rabin-Karp 4 Suffix array/LCP 5 Suffix automaton 6 Compare: 1
KMP/Z: O(n+m); Rabin-Karp expected O(n+m); suffix array implementation is educational and slower than
```

```
1 Dashboard and Statistics
2 Book Management
3 Smart Pattern Search
4 Intelligent Reading Planner
5 Borrowing and Resource Allocation
6 NP-Completeness and Optimization Lab
7 Randomized Algorithms
8 Parallel Algorithms
9 Algorithm Performance Comparison
10 Save Data
0 Exit
Select option: 7
1 Randomized QuickSort 2 Miller-Rabin: 2
Number: 31428
Rounds: 4
Probably prime = false; theoretical false-positive bound 4^-rounds under independent uniform bases.
```

```
1 Dashboard and Statistics
2 Book Management
3 Smart Pattern Search
4 Intelligent Reading Planner
5 Borrowing and Resource Allocation
6 NP-Completeness and Optimization Lab
7 Randomized Algorithms
8 Parallel Algorithms
9 Algorithm Performance Comparison
10 Save Data
0 Exit
Select option: 9
Text: sujith
Pattern: KMP
KMP                   15501 ns        6 comparisons true
Z-function           179900 ns        9 comparisons true
Rabin-Karp         34282501 ns        4 hash checks true
```

```
1 Dashboard and Statistics
2 Book Management
3 Smart Pattern Search
4 Intelligent Reading Planner
5 Borrowing and Resource Allocation
6 NP-Completeness and Optimization Lab
7 Randomized Algorithms
8 Parallel Algorithms
9 Algorithm Performance Comparison
10 Save Data
0 Exit
Select option: 9
Text: algorithm pattern matching
Pattern: pattern
KMP                 197500 ns       26 comparisons true
Z-function          236801 ns       40 comparisons true
Rabin-Karp           41000 ns       20 hash checks true
```